

# Telehealth Consent Assistant: Prototype and UX Insights

Nina C Hubig

IT:U

Linz, Austria

nina.hubig@it-u.at

## Abstract

Telehealth has the potential to improve healthcare accessibility, particularly for individuals in remote or under-served areas. However, its adoption is hindered by barriers in the onboarding process, especially the complexity of legal consent forms, which patients often find difficult to understand or may ignore altogether. To address this, this paper presents an interactive prototype that integrates a specialized artificial intelligence (AI) assistant to strengthen patient understanding of the consent process. The system combines a full-stack web application with a self-hosted, fine-tuned Large language model (LLM), ensuring data privacy and domain-specific behavior. It uses a hybrid approach: the LLM is fine-tuned on telehealth-specific question-answer pairs and enhanced with local context retrieval and structured prompt engineering. A text-to-speech (TTS) component provides audio support, improving accessibility. A user study with 60 participants compared the AI-assisted consent process against a conventional PDF form. Results indicate an improvement in comprehension, enhanced user experience, and increased willingness to use AI-assisted methods. This work demonstrates that a privacy-preserving, AI-enhanced consent assistant can support patient understanding in telehealth. Future research should focus on improving system robustness, refining interaction models, and developing more realistic comprehension assessments to support broader telemedicine adoption.

## CCS Concepts

• **Human-centered computing** → **User interface management systems**; *Visualization*.

## Keywords

Consent, AI-based, Avatar, Telehealth

## ACM Reference Format:

Nina C Hubig. 2026. Telehealth Consent Assistant: Prototype and UX Insights. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798990>

## 1 Introduction

Telehealth is the practice of delivering healthcare services via digital means such as video calls. It emerged especially during the COVID-19 pandemic, with the promise of making healthcare more accessible - particularly for those living in remote or under-served

areas [4, 11]. However, despite its potential benefits, the adoption of telehealth still faces significant barriers. Receiving healthcare services through digital platforms carries a social stigma, which can affect patients' willingness to engage in this form of treatment [11]. Another hurdle patients are struggling with is the onboarding process, in which they must consent to legal terms. This process is often considered complex and cumbersome, and the legal jargon is viewed as difficult to understand, posing an accessibility barrier that may dissuade potential patients from participating in this form of healthcare, and, as potential consequence, from accessing healthcare altogether [8]. To address this gap, this extended abstract presents a General Data Protection Regulation (GDPR)-compliant telehealth consent assistant that enhances patient understanding while ensuring data sovereignty. The system employs a hybrid AI architecture: a fine-tuned LLM is enhanced with local context retrieval and structured prompt engineering, enabling accurate, context-aware explanations without relying on thirdparty services. A Text-to-speech (TTS) component further improves accessibility [2, 8]. This leads to the research question of this study: Do AI-enhanced consent assistants improve a patient's capacity to consent, their understanding of consent forms, and perceived ease of use compared to standard consent form processes, and how do prior experiences with consent forms or healthcare AI influence these outcomes? To evaluate the usefulness and acceptance of this approach, a user study was conducted with a group of 60 participants, divided into two subgroups. The first group used the AI consent assistant; the second group interacted with a standard PDF form. The study assessed participants' capacity to consent, their user experience, and their willingness to use AI in the context of healthcare. The user study employed a mixed-methods approach, combining quantitative measures including capacity to consent assessments and standardized user experience questionnaires-with qualitative analyses of open-ended participant responses. This approach aims to provide a comprehensive evaluation of how an AI-enhanced method influences both, objective comprehension and subjective user perceptions [1, 9, 12].

## 2 Related Work

The integration of artificial intelligence (AI) into healthcare workflows has expanded rapidly, aiming to improve patient engagement, reduce administrative burden, and enhance informed consent processes [8, 13]. AI in digital consent is a promising application to address the complexity and legal challenges inherent in patient consent, particularly in telehealth. These tools aim to simplify consent language, improve comprehension, and streamline form completion while maintaining legal correctness. This section reviews relevant work in AI for healthcare, with a focus on privacy-preserving adaptation, human-AI interaction, and the broader landscape of digital consent.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798990>

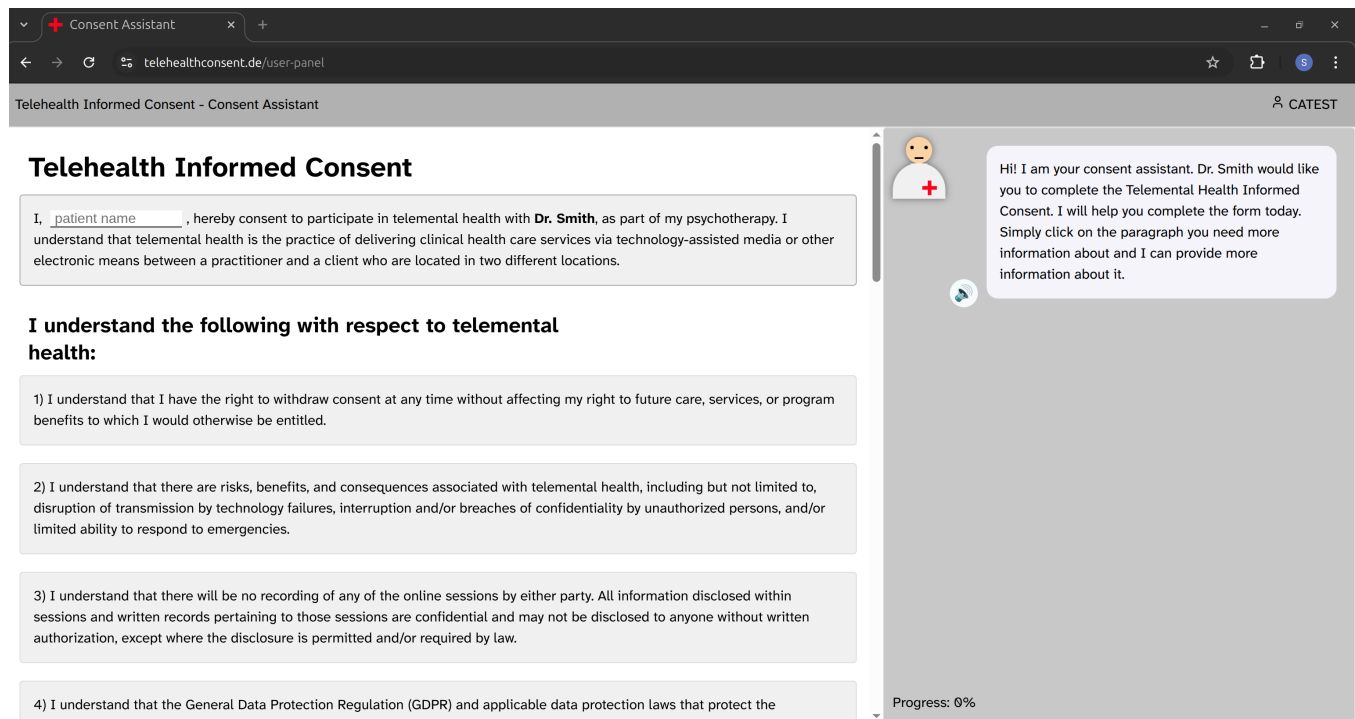


Figure 1: Telehealth Consent Assistant: UI with form on the left and AI assistant on the right.

## 2.1 AI-Enhanced Consent and User Experience

Studies demonstrate that interactive consent assistants using Natural language processing (NLP) can improve comprehension and reduce cognitive load by translating complex legal documents into accessible language [8, 13]. These systems often employ chatbots or voice interfaces to guide users through consent terms, with evidence showing improved usability and reduced mental workload [10]. However, achieving legally adequate comprehension remains a challenge, particularly for users with low health literacy or limited digital access.

A 2025 scoping review highlights both the promise and risks of AI in digital consent [8]. While AI can improve comprehension and efficiency, it may introduce bias, automation bias, and exclusion of vulnerable populations. The review emphasizes that AI tools should complement, not replace, human-led consent discussions—a principle aligned with this work, which supports rather than supplants patient agency.

Usability is typically measured using standardized metrics such as the System Usability Scale (SUS), NASA Task Load Index (NASA-TLX), and Net Promoter Score (NPS) [5, 9]. The Technology Acceptance Model (TAM) further informs evaluation by assessing perceived usefulness and ease of use [6]. These tools provide a multidimensional view of user experience, essential for evaluating AI in sensitive healthcare contexts.

While prior studies have shown that AI-based consent assistants can improve comprehension and reduce cognitive load [8], this work advances these efforts by leveraging a fine-tuned LLM enhanced with retrieval-augmented local context and a TTS interface

to further improve accessibility and understanding. The presented system balances legal rigor and user experience by operating fully within a GDPR-compliant environment, addressing privacy concerns highlighted in earlier work.

## 2.2 AI Avatars in Healthcare

The use of an animated 2D avatar in this system is grounded in research on human-AI interaction. Studies show that anthropomorphic interfaces improve patient comprehension, reduce anxiety, and encourage self-disclosure, particularly among users with low health literacy [7]. Avatars provide a sense of continuity and empathy, making digital interactions feel more personal and trustworthy.

Research indicates that patients disclose more personal health information to AI avatars than to text-based chatbots, especially when the avatar exhibits human-like responsiveness [14]. Even simple avatars can enhance perceived empathy and usability in digital consent processes, making them a practical choice for real-world deployment [7].

However, research also emphasizes important challenges and ethical considerations related to AI avatars in healthcare. Almomani et al. (2025) discuss foundational ethical concerns such as privacy risks arising from handling sensitive personal health data, the necessity for innovation balanced with user privacy, and user trust in AI-driven avatars in immersive environments. Moriuchi (2022) highlights how biases in AI systems can influence avatar behavior and decision-making, underscoring the need for transparency and preserving patient autonomy through explainable interactions. Additionally, cultural sensitivity and user customization are critical for

acceptance, as avatars must be designed to avoid alienating users via the uncanny valley effect or cultural mismatch [3, 15].

Emerging work in avatar technology points towards extended reality (XR) and 3D immersive avatars as promising future directions. Šarić et al. (2023) present a telemedicine collaboration system that uses patient avatars with 3D body pose estimation, illustrating advances in realistic, interactive avatar design for healthcare applications. Ong et al. (2021) review how XR technologies have enhanced telehealth capabilities during and beyond the COVID-19 pandemic, enabling more immersive and context-aware patient interactions. These innovations integrate AI avatars with real-time sensor data and natural language understanding, paving the way for emotionally intelligent, adaptive healthcare companions that can support telemedicine, rehabilitation, and mental health interventions [16, 17].

### 3 Prototype Design

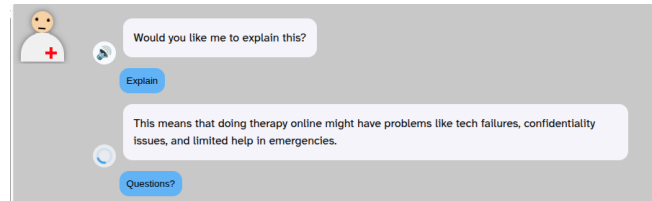
The user interface is the primary point of contact between the patient and the consent assistant. It is designed to be intuitive, accessible, and privacy-preserving. To support patients in understanding the legal terms and guide their interactions, the interface is divided into two main sections: the form interface (left) and the AI assistant interface (right). The form interface presents the legal document separated into logical sections which interactively activate the AI assistant. The AI assistant provides explanations and a chat interface aiding understanding of both the interface and the legal sections, as well as an audio version of the interactions. This layout allows users to focus on the consent document while maintaining seamless access to AI support.

The form interface displays the telehealth consent document as a sequence of interactive paragraph tiles, each containing a section of the legal text and associated input fields for personal information (e.g. name, date of birth, contact details). Users can scroll through the form and fill in required fields at their own pace. To request an explanation, the user clicks on a paragraph to select it, which is visually represented by a color highlight; this interaction activates the consent assistant interface. At the end of the form, a signature field is displayed which only becomes active once all required fields are filled. A progress indicator in the assistant panel shows the percentage of completed fields.

The consent assistant is represented by an animated 2D avatar styled as a friendly cartoon doctor (Figs. 1 and 2). The avatar provides idle animations (e.g. blinking, subtle head and shoulder movements) and synchronized mouth movements during speech, creating a more engaging and approachable interaction. When a paragraph is selected, the assistant interface offers two actions: *Explain*, which generates a concise textual explanation of the selected paragraph, and *Questions?*, which opens a free-form chat interface for in-depth inquiries (Fig. 2). Each speech bubble is accompanied by an audio playback button, allowing users to listen to the TTS-generated audio.

### 4 Method

To evaluate the user experience of this interface, a between-groups study compared the AI-powered consent assistant (Consent Assistant (CA) group,  $n = 29$ ) with a standard PDF-based consent



**Figure 2: Avatar UI with idle animations and interface with explaining functions: "Explain" and "Questions?" are action buttons that allow for in-depth AI-generated explanations**

form (Web Form (WB) group,  $n = 30$ ). Both groups completed the consent task within a five-minute time frame and then filled out standardized user experience questionnaires: SUS for usability, NASA-TLX for cognitive workload, TAM for acceptance, and NPS for satisfaction and likelihood to recommend.

### 5 Results

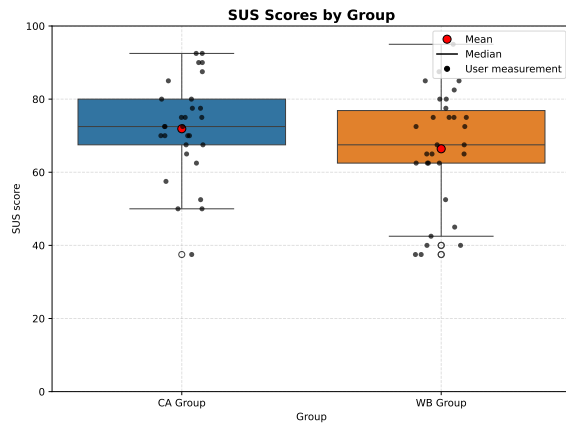
The SUS revealed higher perceived usability for the consent assistant (Mean = 71.90) compared to the web-based form (Mean = 66.42), indicating a moderate usability advantage for the interactive UI. The NASA-TLX scores showed significantly lower perceived mental workload in the CA group (Mean = 2.16) than in the WB group (Mean = 2.64; Mann-Whitney U,  $p = 0.0068$ ), suggesting that the avatar-based assistant and contextual explanations reduced cognitive strain.

TAM results showed similarly high perceived usefulness in both conditions, but perceived ease of use was significantly higher for the consent assistant (Mean = 5.60) than for the web form (Mean = 4.89; Mann-Whitney U,  $p = 0.0028$ ). This indicates that the combination of paragraph-based explanations, avatar guidance, and progress feedback made the interaction feel easier and more controllable.

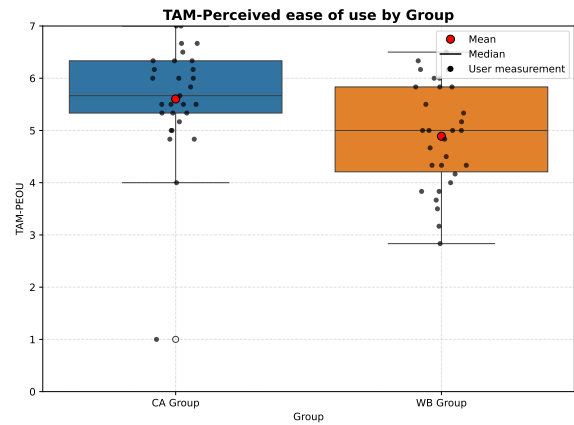
Despite these advantages, NPS scores were negative in both groups (CA: -24.14; WB: -16.67), reflecting considerable hesitation to recommend either consent method. Open-ended feedback highlighted usability and design suggestions as dominant themes, with many participants appreciating the explanatory support of the assistant but also pointing out performance issues and the novelty of AI in consent contexts. Overall, the evaluation suggests that the consent assistant interface improves ease of use and reduces cognitive workload compared to a standard web form, while also revealing important opportunities for further refinement of trust, responsiveness, and satisfaction.

### 6 Discussion

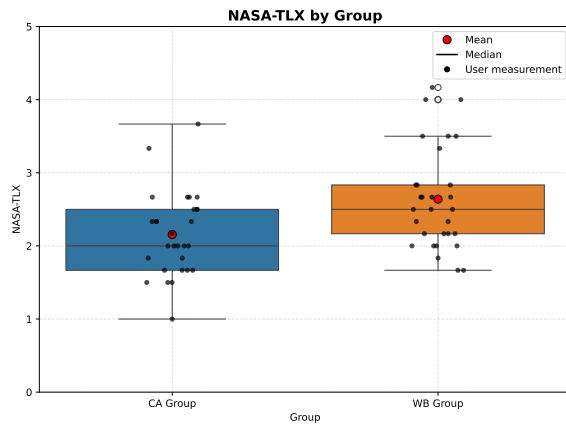
The comprehensive evaluation of user experience across multiple metrics - including the System Usability Scale (SUS), NASA Task Load Index (NASA-TLX), Technology Acceptance Model (TAM), and Net Promoter Score (NPS) - consistently demonstrated that the AI-enhanced consent system surpasses the traditional web form in perceived usability, perceived ease of use, and mental workload reduction. SUS scores placed the AI tool above average, indicating a generally positive perception, while NASA-TLX results reflected a moderate decrease in perceived cognitive effort. TAM analysis revealed that both methods were highly rated for usefulness. However,



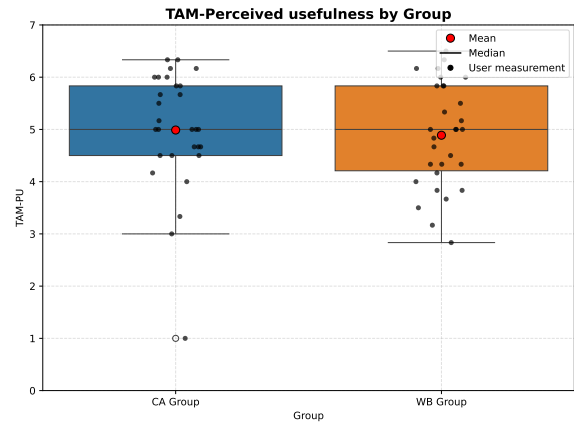
(a) SUS scores comparing the AI-assisted consent tool (CA) and control system (WB).



(a) TAM perceived ease of use per group.



(b) NASA-TLX scores: lower scores indicate lower perceived workload. The CA group reported reduced workload compared to the WB group.



(b) TAM perceived usefulness by group

the AI system was favored for ease of use, suggesting it requires less user effort to operate. Nonetheless, the overall negative NPS scores in both groups highlight a layer of dissatisfaction that warrants attention. This could emanate from the novelty and inherent complexity of AI systems in healthcare settings, as well as unfulfilled participant expectations. Moreover, these metrics, though valuable, provide a multidimensional perspective-integrating usability, workload, acceptance, and satisfaction-that highlights strengths and areas for improvement more holistically. Incorporating qualitative feedback from participants open-ended responses further strengthens this interpretation by revealing specific issues around trust, interface design, perceived redundancy, and overall engagement, which these quantitative measures alone cannot fully capture. It is important to acknowledge the limitations of each metric: NASA-TLX's complexity might burden participants in less demanding scenarios, and NPS's sensitivity to context could influence its scores. Therefore, a balanced, critical interpretation suggests that while Consent assistance in consent procedures holds promise, systematic enhancements are necessary. These should focus on improving

AI performance, transparency, and user trust, alongside refining interface design. Future research should continue to employ a synergistic approach, combining objective usability, subjective workload, and qualitative data, to optimize AI-enabled healthcare tools and foster broader user acceptance.

## 7 Future Work and Conclusion

Future iterations could upgrade the 2D comic avatar to realistic 3D models with synchronized gestures to enhance trust and immersion. The current split-screen interface requires redesign for mobile responsiveness, potentially as a native smartphone app. Voice-first interaction via speech-to-text could replace manual clicks, better serving users with low literacy. The system should be evaluated with diverse cohorts using complex clinical trial consent forms to demonstrate impact beyond simplified telehealth scenarios. Finally, the Kokoro TTS must be replaced with an edge-optimized model to eliminate performance bottlenecks observed during extended use. Alternatively to a 3D expansion a switch to a XR could be an interesting research avenue in which a speech (by the user) to speech (by the AI assistant) scenario could be evaluated in comparison to the current state of text given consent.

Overall, this work contributes empirical evidence that AI-supported consent interfaces can meaningfully improve the user experience of telehealth consent, while also revealing critical design and trust-related challenges that remain unresolved. By adopting a holistic evaluation strategy, the study highlights both the potential and the constraints of deploying conversational AI systems in ethically sensitive healthcare workflows.

## References

- [1] Corey Adams, Ramesh Walpola, Anthony M Schembri, and Reema Harrison. 2022. The ultimate question? Evaluating the use of Net Promoter Score in healthcare: A systematic review. *Health Expectations* 25, 5 (2022), 2328–2339.
- [2] Rohaid Ali, Ian D. Connolly, Oliver Y. Tang, Fatima N. Mirza, Benjamin Johnston, Hael F. Abdulrazeq, Rachel K. Lim, Paul F. Galamaga, Tiffany J. Libby, Neel R. Sodha, Michael W. Groff, Ziya L. Gokaslan, Albert E. Telfeian, John H. Shin, Wael F. Asaad, James Zou, and Curtis E. Doberstein. 2024. Bridging the literacy gap for surgical consents: an AI-human expert collaborative approach. *npj Digital Medicine* 7, 1 (2024), 63. doi:10.1038/s41746-024-01039-2
- [3] Ammar Almomani, Ahmad Alââşqerem, Mohammad Alauthman, Amjad Aldweesh, Samer Aoudi, and Said A Salloum. 2025. Ethical Foundations of AI-Driven Avatars in the Metaverse for Innovation and User Privacy. *IEEE Access* (2025).
- [4] American Hospital Association. 2025. Fact Sheet: Telehealth. *American Hospital Association* (2025). <https://www.aha.org/fact-sheets/2025-02-07-fact-sheet-telehealth> Accessed September 2025.
- [5] John Brooke. 1986. *SUS: A Quick and Dirty Usability Scale*. Technical Report TR/86/100. Digital Equipment Corporation, Reading, UK.
- [6] Fred D Davis. 1989. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS quarterly* (1989), 319–340.
- [7] Arno Joachim Gingele, Hesam Amin, Aniek Vaassen, Ivana Schnur, Cathy Pearl, Hans-Peter Brunner-La Rocca, and Josiane Boyne. 2023. Integrating avatar technology into a telemedicine application in heart failure patients: A pilot study. *Wiener klinische Wochenschrift* 135, 23 (2023), 680–684.
- [8] Mascha Goldschmitt, Patricia Gleim, Sekina Mandelartz, Philipp Kellmeyer, and Thomas Rigotti. 2025. Digitalizing informed consent in healthcare: a scoping review. *BMC Health Services Research* 25 (2025), 893.
- [9] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology*. Vol. 52. North-Holland, 139–183.
- [10] Yihao Hou, Christian Bert, Ahmed Gomaa, Gitta Lahmer, Daniel Höfler, Tim Weissmann, and Felix Putz. 2025. Fine-tuning a local LLaMA-3 large language model for automated privacy-preserving physician letter generation in radiation oncology. *Frontiers in Artificial Intelligence* 7 (2025), 1493716. doi:10.3389/frai.2024.1493716
- [11] Mackenzie Kemp, Kristin L Rising, Gregory Laynor, Jessica Miao, Brooke Worster, Anna Marie Chang, Andrew J Monick, Amanda Guth, Tracy Esteves Camacho, Kiana McIntosh, et al. 2025. Barriers to telehealth uptake and use: a scoping review. *JAMIA open* 8, 2 (2025), ooaf019.
- [12] James R Lewis. 2018. The system usability scale: past, present, and future. *International Journal of Human-Computer Interaction* 34, 7 (2018), 577–590.
- [13] Subhankar Maity and Manob Jyoti Saikia. 2025. Large Language Models in Healthcare and Medical Applications: A Review. *Bioengineering* 12, 6 (2025), 631.
- [14] Sara Montagna, Stefano Ferretti, Lorenz C. Klopfenstein, Michelangelo Ungolo, Martino F. Pengo, Gianluca Aguzzi, and Matteo Magnini. 2025. Privacy-preserving LLM-based chatbots for hypertensive patient self-management. *Smart Health* 36 (2025), 100552. doi:10.1016/j.smhl.2025.100552
- [15] Emi Moriuchi. 2022. Leveraging the science to understand factors influencing the use of AI-powered avatar in healthcare services. *Journal of Technology in Behavioral Science* 7, 4 (2022), 588–602.
- [16] Triton Ong, Hattie Wilczewski, Samantha R Paige, Hiral Soni, Brandon M Welch, and Brian E Bunnell. 2021. Extended reality for enhanced telehealth during and beyond COVID-19. *JMIR serious games* 9, 3 (2021), e26520.
- [17] Matko Saric, Mladen Russo, Luka Kraljevic, and Davor Meter. 2023. Extended reality telemedicine collaboration system using patient avatar based on 3d body pose estimation. *Sensors* 24, 1 (2023), 27.